Seonghyun Park

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Summary

I'm a second-year Ph.D. student at Graduate School of AI at KAIST, advised by Sungsoo Ahn. My research interests lie mainly in **AI4Science**, and I believe that machine learning can accelerate scientific discovery. Recently, I have been focusing on incorporating experimental data into models to better understand protein-ligand interactions.

Education

Ph.D. **Korea Advanced Institute of Science and Technology (KAIST)** — Seoul, South Korea
Kim Jaechul Graduate School of Artificial Intelligence — Feb 2025 – present
- Structured and Probabilistic Machine Learning (SPML) Lab @ Sungsoo Ahn
- Protein-drug interactions, Molecular dynamics (MD)

M.S. **Pohang University of Science and Technology (POSTECH)** — Pohang, South Korea
Graduate School of Computer Science and Engineering (CSE) — Mar 2023 – Feb 2025
- Machine Learning Lab @ Sungsoo Ahn
- Graph Neural Networks (GNNs), Over-squashing

E.S. **Institut National des Sciences Appliquées (INSA) Lyon** — Lyon, France
Bioinformatique, exchange student — Jan 2022 – June 2022

B.S. **Pohang University of Science and Technology (POSTECH)** — Pohang, South Korea
Computer Science and Engineering (CSE) — Feb 2019 – Feb 2023

Experience

Machine Learning Lab, Research intern — Pohang, South Korea
Graph Neural Networks (GNNs), advised by Sungsoo Ahn — July 2022 – Jan 2023

Bagelcode, Business Analyst (BA) intern — Seoul, South Korea
Game economy management and KPI analysis automation — June 2021 – Aug 2021

Seller Hub, Product Manager (PM) intern — Seoul, South Korea
Organization-wide task prioritization and landing funnel renewal — July 2020 – Aug 2020

Publication

† Equal contribution · ‡ Co-corresponding author

Blending experimental electron density information to protein-ligand interactions
Seonghyun Park, Shuan Chen, Jun Jin Choong, Noboru Kuno, Gyubok Lee, Kaushalya Madhawa, Keisuke Ozawa, Hikaru Shindo, Sungsoo Ahn
NeurIPS 2026 Workshop, AI4DD

A Systematic Evaluation of Co-folding Model Representations for Small-Molecule Learning
Hyosoon Jang, Hyunjin Seo, Honghui Kim, ***Seonghyun Park***, Taewon Kim, Yunhui Jang, Sungsoo Ahn
NeurIPS 2026 (10.48550/arXiv.2602.13249)

INDIBATOR: Diverse and Fact-Grounded Individuality for Multi-Agent Debate in Molecular Discovery
Yunhui Jang, ***Seonghyun Park***, Jaehyung Kim, Sungsoo Ahn
ICML 2026 Workshop, AI4Science (10.48550/arXiv.2602.01815)

Riemannian MeanFlow
Dongyeop Woo, Marta Skreta, ***Seonghyun Park***, Kirill Neklyudov, Sungsoo Ahn
ICML 2026 (10.48550/arXiv.2602.07744)

Learning Collective Variables from BioEmu with Time-lagged Generation
Seonghyun Park, Kiyoung Seong, Soojung Yang, Rafael Gomez-Bombarelli, Sungsoo Ahn
ICLR 2026 (10.48550/arXiv.2507.07390)

Transition Path Sampling with Improved Off-Policy Training of Diffusion Path Samplers
Kiyoung Seong, ***Seonghyun Park***, Seonghwan Kim, Woo Youn Kim, Sungsoo Ahn
ICLR 2025 (10.48550/arXiv.2405.19961)

Non-backtracking Graph Neural Networks
Seonghyun Park[†], Narae Ryu[†], Gahee Kim, Dongyeop Woo, Se-Young Yun[‡], Sungsoo Ahn[‡]
TMLR 2024, NeurIPS 2023 Workshop GLFrontiers Oral (10.48550/arXiv.2310.07430)

Diffusion Probabilistic Models for Structured Node Classification
Hyosoon Jang, ***Seonghyun Park***, Sangwoo Mo, Sungsoo Ahn
NeurIPS 2023 (10.48550/arXiv.2302.10506)